

The “Oracle of Anime”: Finding the Anime Industry’s Kevin Bacon

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This study analyzes the Japanese voice-acting industry by modeling the top 10,000 anime titles on MyAnimeList as a unweighted unipartite collaboration network. Unique voice actors (“seiyuu”) were treated as nodes, with shared anime credits forming edges. Using Closeness Centrality, shortest-path analysis, and Small World validation, the research identifies the anime industry’s “Kevin Bacon” and evaluates its structural interconnectedness. The final network consisted of 4,614 seiyuu and 470,639 collaboration edges, with 99.8% of actors belonging to a single giant component. Results identified Sawashiro Miyuki as the most central seiyuu based on Closeness Centrality. The network exhibited a high clustering coefficient ($C = 0.7400$) and a low average path length ($L = 2.24$), closely approximating a random graph’s efficiency while maintaining significantly stronger local clustering. These findings confirm that the anime voice-acting industry mathematically qualifies as a Small World network, where tightly knit professional communities are linked through highly central collaborative hubs. This study provides a quantitative framework for understanding talent distribution and professional connectivity within one of Japan’s largest cultural industries.

Keywords: Social Networks, Small World, Kevin Bacon, Anime, Seiyuu

I. INTRODUCTION

The concept of “Small Worlds” was first popularized by social psychologist Stanley Milgram following his publication of “The Small World Problem” in *Psychology Today* [1]. In his experiments, Milgram asked a random group of Americans living in Kansas and Nebraska to send a letter to a stockbroker in Boston. The catch—participants could only send letters to people they were on a first-name basis with, and thus should mail it to someone they believe would be “closer” or more likely to know the stockbroker personally. The recipients of the letters should also do the same. Surprisingly, Milgram found that not only were most letters able to reach the stockbroker, they were able to do so in a very short chain of intermediary links, the average of which was six, hence the phrase “six degrees of separation”. Milgram referred to this unexpected shortness of distance between people as “the small world problem”.

This idea of the “Small World” phenomenon is seen again in the “Six Degrees of Kevin Bacon” game [2]. The game first started when three students appeared on a television talk show and demonstrated that Kevin Bacon could be linked to every known Hollywood actor or actress through a series of shared credits with their colleagues. For example, Sean Connery and Bacon have never played in a movie together. However, Connery played in *The Untouchables* with Stephen Burrows who played in *She’s Having a Baby* with Bacon; therefore, Connery’s Bacon-number is 2, indicating two degrees of separation from Bacon himself. Two computer scientists used the Internet Movie Database to set up a web version of this game ¹ and proved that most actors and actresses in the movie industry had small Bacon-numbers, illus-

trating the interconnectedness of Hollywood.

The primary goal of this project is to identify the most “central” Japanese voice actor (“seiyuu”) in the anime industry and determine if the “Small World” phenomenon [3] exists within this professional community. Just as the “Six Degrees of Kevin Bacon” game highlights the interconnectedness of Hollywood, this study seeks to find the center of the anime industry using a dataset ² of the top 10,000 anime titles on MyAnimeList ³.

The core research questions are:

- Who is the ‘Kevin Bacon’ of the anime industry based on Closeness Centrality?
- What is the average ‘degree of separation’ between any two seiyuus in the top 10,000 anime?
- Does the Seiyuu collaboration network exhibit ‘Small World’ characteristics, specifically high local clustering and short average path lengths?

Understanding these connections is important because it reveals the structural backbone of the industry. By quantifying the ‘reach’ of specific talent, the study sets up a framework for understanding how professional collaborations link disparate parts of the animation ecosystem.

¹ <https://oracleofbacon.org/>

² <https://www.kaggle.com/datasets/furkanark/myanimelist-top-10000-anime-dataset>

³ <https://myanimelist.net/>

II. RELATED WORKS

A. Small World Phenomenon and Mathematical Proof

The theoretical foundation of this study is rooted in the work of Watts and Strogatz, who formalized and mathematically proved the “Small-World” phenomenon [3]. They posed that many real-world networks are neither perfectly ordered nor entirely random; instead, they exhibit a unique topology characterized by a high clustering coefficient (C) and a low average path length (L). In this context, a high clustering coefficient indicates that a seiyuu’s colleagues are likely to have worked with one another, forming professional “cliques,” while a low path length ensures that any two seiyuus in the industry can be connected through a remarkably short chain of shared credits, or the “six degrees of separation”.

Watts and Strogatz also claimed that a network is only a “Small World” if its path length is comparable to a random graph ($L \approx L_{rand}$) but its clustering is significantly higher ($C \gg C_{rand}$). This definition is used later to validate the existence of the “Small World” phenomenon in the anime industry to prove that the interconnectedness of the seiyuu ecosystem is a product of deliberate professional structure rather than a statistical fluke.

B. Centrality Metrics

To identify the industry’s most influential figures, I applied the framework established by Freeman regarding Centrality [4]. While Freeman defined several ways to measure a node’s importance (Degree, Betweenness, and Closeness), this project prioritized Closeness Centrality. Mathematically defined as the inverse of the sum of the shortest distances between a node and all other nodes in the network, Closeness Centrality is the metric for identifying our “Kevin Bacon” figure. It measures how “independent” or “reachable” an actor is, quantifying their ability to spread influence or access any other part of the network with maximum efficiency. I also considered Degree Centrality (total number of unique collaborators) and Betweenness Centrality (the frequency with which an actor sits on the shortest path between others) to identify “gatekeepers” who bridge disconnected portions of the network.

C. Precedent in Collaboration Networks

While substantial research has been conducted on Western media and academic circles, there is a notable gap in the literature regarding a large-scale mapping of the Japanese anime industry. This study builds upon the methodology of Newman [5], who applied network science to co-authorship databases in physics and biology. Newman’s work demonstrated that collaboration

networks naturally gravitate toward a “Giant Component”—a single, massive interconnected web that contains the vast majority of participants.

This study adapts Newman’s findings to the Seiyuu ecosystem, expecting to find a similar Giant Component that links decades of production history. While Newman focused on the “gatekeeping” role of scientists, I will use his concepts of community detection to see how a singular “Kevin Bacon” actor facilitates the flow of talent across a media landscape that spans diverse genres and studios. By identifying this central nexus, I provide the first quantitative assessment of the structural robustness of one of Japan’s most significant cultural exports.

III. METHODS

A. Data Acquisition and Preprocessing

Mapping the entire industry across all time would require an extortionate amount of computing power, hence why I limited the scope of this project. The “MyAnimeList Top 10,000 Anime” dataset from Kaggle is up to date as of the year this paper is written. It also limits the scope of the study to the top 10,000 titles to ensure that the $O(V \cdot E)$ complexity of the Closeness Centrality and All-Pairs Shortest Path calculations remains computationally feasible within a standard local development environment. This dataset also contains structured csv files for anime titles (`anime.csv`) mapped to their respective voice casts (`anime_voice_actors.csv` and `anime_characters.csv`), making it incredibly easy to map each Seiyuu with their colleagues.

The preprocessing pipeline follows a relational join strategy using Python’s Pandas library to ensure data integrity. By merging the `anime.csv`, `anime_voice_actors.csv` and `anime_characters.csv` tables on their respective foreign keys using the `entities.csv` file, the data transformed into a normalized long-form feature set. To maintain the study’s focus on the Japanese industry core, the dataset is filtered to include only ‘Japanese’ language credits. Finally, I dropped all duplicate entries and only kept the anime ID, anime title and seiyuu name columns to form a table (`merged_df`) that tracks every seiyuu’s voice acting credits. There were 152952 unique acting credits in total.

B. Network Modeling

While the original data is bipartite (Seiyuu $\leftrightarrow$ Anime), for my goal of Closeness Centrality analysis, an unweighted unipartite graph approach is much simpler and enough. In this model, every unique Seiyuu is a node. An edge (connection) is created between two nodes if both seiyuus appear in the credits of the same anime title.

Using Python’s NetworkX library, edges were generated by grouping actors by anime title in the `merged_df`

table and constructing pairwise combinations among all actors within each production. This process produced a collaboration graph containing 4,614 unique seiyuu and 470,639 edges.

C. Network Analysis

To answer the research questions, the following were NetworkX functions were used:

- `nx.closeness centrality()` to calculate the closeness centrality of all actors. The seiyuu with the highest score was crowned the ‘Kevin Bacon’ of Anime.
- `nx.betweenness centrality()` to identify the ‘bridges’ within the network. While not the primary metric for the Kevin Bacon title, betweenness allows us to quantify which seiyuus are most responsible for facilitating the ‘Small World’ characteristics (specifically the short average path lengths) observed in the industry.
- `nx.average_shortest_path_length()` to calculate the average shortest path between all pairs of nodes. This was necessary to find the ‘average degrees of separation’.
- `nx.average_clustering()` to measure the clustering coefficient and determine how often an seiyuu’s co-stars also work with each other, confirming the ‘Small World’ nature of the industry. A score close to 1 suggests a very tight-knit industry.

Since some of these methods require a fully connected graph, in the case of a disconnected graph, I isolated the Largested Connected Component (LCC) of the graph by using `nx.connected_components(G)` to find the biggest connected component, then applied the above methods to the LCC.

D. Evaluation

To validate the Small World hypothesis, the analysis focused on the network’s Largest Connected Component (LCC), which contained 4,604 actors (99.8% of all nodes).

The empirical network’s average clustering coefficient and average shortest path length were compared to those of an Erdős–Rényi random graph with equivalent size and density. Small World characteristics were confirmed if:

- Empirical clustering coefficient was substantially larger than the random model
- Empirical average path length remained comparable to the random model

Additionally, shortest-path distributions from the most central actor were visualized to confirm the practical “degrees of separation” structure across the industry.

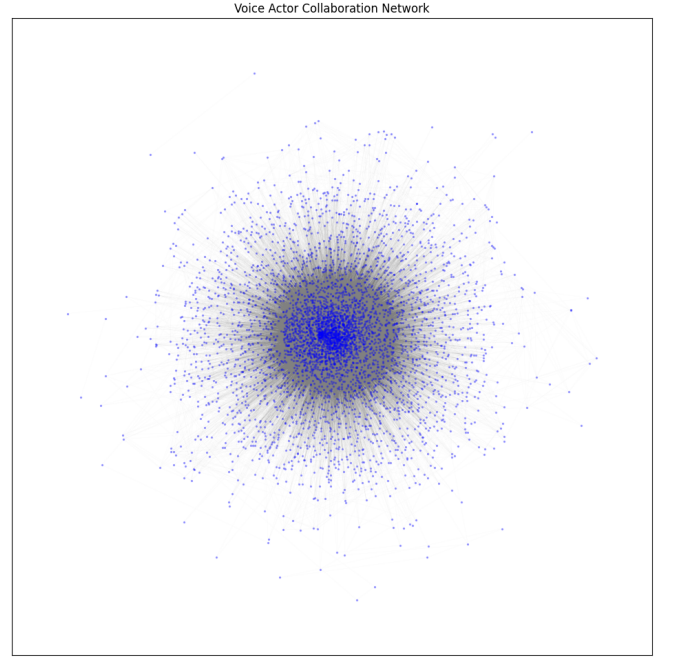


Figure 1. Seiyuu collaboration network (4614 seiyuus and 470639 connections)

IV. RESULTS

The network modeling phase successfully mapped the professional connections within the top 10,000 anime titles. After processing approximately 152,952 unique voice-acting credits from the MyAnimeList dataset, a projected unipartite graph was constructed consisting of 4,614 nodes (unique seiyuu) and 470,639 edges (shared project credits).

A. Identification of the ‘Kevin Bacon’ of Anime

To answer the primary research question, Closeness Centrality was calculated for every seiyuu in the network. The analysis identified Sawashiro Miyuki as the seiyuu with the highest closeness centrality score (0.6300), and identified them as the “Kevin Bacon” of the anime industry. This finding suggests that Sawashiro is the most “reachable” figure in the professional landscape, facilitating the shortest average path to any other seiyuu within the top 10,000 titles.

B. Average Degrees of Separation

The “Small World” hypothesis was tested by calculating the average shortest path length (L) across the Largest Connected Component (LCC) of the graph (4,604 nodes, which is 99.8% of the nodes of the original entire graph). The empirical results reveal an average

Table I. Top 10 Seiyuus by Closeness Centrality

Seiyuu	Closeness Centrality
Sawashiro, Miyuki	0.6300
Sakurai, Takahiro	0.6252
Sugita, Tomokazu	0.6214
Fukuyama, Jun	0.6193
Hanazawa, Kana	0.6170
Inoue, Kikuko	0.6145
Yamadera, Kouichi	0.6141
Sakamoto, Maaya	0.6129
Ootsuka, Akio	0.6111
Morikawa, Toshiyuki	0.6099

degree of separation of 2.24. This indicates that any two voice seiyuus in the modern anime industry can typically be linked by just over two shared projects.

Table II. Small World Validation (Average Clustering Coefficient C and Average Shortest Path Length L)

Network Type	C	L
Empirical (Anime)	0.7400	2.24
Random (Control)	0.0444	1.96

C. Validation of Small World Characteristics

Following the methodology of Watts and Strogatz, the seiyuu network was compared to an Erdős-Rényi random graph of identical size ($n = 4614, m = 470639$):

- Clustering Coefficient (C): The seiyuu network exhibited a coefficient of 0.7400, whereas the random model yielded 0.0444.
- Average Path Length (L): The seiyuu network length of 2.24 was comparable to the random model's 1.96.

Because $C \gg C_{rand}$ and $L \approx L_{rand}$, the Japanese voice-acting industry mathematically qualifies as a “Small World” network.

D. Distribution of Degrees of Separation from Anime’s ‘Kevin Bacon’

To further evaluate the practical structure of the collaboration network, shortest-path distances were calculated from Sawashiro Miyuki to every other seiyuu within the LCC. The resulting distribution strongly reinforces the Small World nature of the anime industry.

The analysis revealed that:

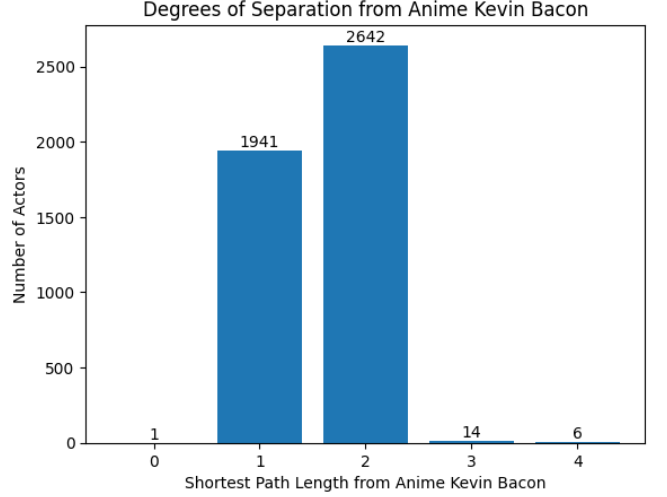


Figure 2. Distribution of shortest paths to Sawashiro Miyuki

- 1 actor (Sawashiro Miyuki) had a path length of 0
- 1,941 actors were directly connected (distance = 1)
- 2,642 actors were connected through two intermediaries (distance = 2)
- Only 14 actors required three degrees of separation
- Just 6 actors required four degrees of separation

These findings demonstrate that the overwhelming majority of seiyuus (over 99%) are within one or two collaborative steps of the network’s central hub. The concentration of actors at distances one and two highlights an exceptionally compressed professional ecosystem, where even distant collaborators remain remarkably accessible.

This distribution provides strong empirical support for the closeness centrality findings, confirming that Sawashiro Miyuki functions as a highly efficient connector across the industry. More broadly, the extremely short separation distances illustrate how a small number of influential hubs maintain global cohesion across thousands of professionals, solidifying the anime voice-acting network as a mathematically robust Small World system.

V. DISCUSSION

The results confirm that the seiyuu industry is an exceptionally tight-knit ecosystem. The high local clustering ($C = 0.7400$) suggests that professional cliques are highly prevalent; if Seiyuu A works with Seiyuu B and Seiyuu C, there is a 74% probability that B and C have also worked together.

The dominance of seiyuu like Sawashiro Miyuki is mathematically evidenced by their high Closeness Centrality scores, which identify them as the most efficient

connectors within the top 10,000 anime titles. By maintaining the shortest average path lengths to all other nodes, these seiyuus function as “super-hubs”. These hubs effectively shrink the industry’s professional distance by serving as common links between disparate clusters. Their presence is the primary reason the average degree of separation remains as low as 2.16 across the entire dataset.

The structural robustness of this Small World network ensures that the industry remains highly interconnected. With a Clustering Coefficient of 0.7400, the network is characterized by dense local cliques where seiyuus frequently collaborate within the same circles. The data indicates that the industry is efficient at moving information or standards through these central hubs.

The presence of a 99.8% Giant Component further emphasizes the extraordinary structural integration of the anime voice-acting industry. With only a negligible fraction of actors existing outside the primary network, the seiyuu ecosystem functions as an almost entirely unified professional web. This level of connectivity suggests that major actors such as Sawashiro Miyuki not only serve as central hubs but also bridge multiple generations, genres, and production circles, reinforcing both the resilience and collaborative efficiency of the industry. However, the Giant Component also suggests that while the industry is efficient, it may also be difficult for new, unlinked talent to penetrate the central core of high-closeness professionals.

A. Limitations

The current model utilizes a projected unipartite graph with unweighted edges. This approach may create ‘extra’ connections between seiyuus who only worked together once, potentially hiding the nuanced hierarchy of professional partnerships. Furthermore, the results are limited to the top 10,000 most popular titles, which may skew the network toward modern production eras and commercially successful talent.

VI. CONCLUSION

This study successfully mapped the Japanese voice-acting industry as a large-scale collaboration network and demonstrated that it strongly exhibits Small World characteristics. By modeling 4,614 voice actors across the

top 10,000 anime titles, the analysis identified Sawashiro Miyuki as the “Kevin Bacon” of anime based on the highest Closeness Centrality score, indicating the most efficient professional reach across the industry.

The network’s structure revealed an average path length of only 2.24 degrees of separation within its Largest Connected Component, confirming that nearly all major seiyuu are connected through remarkably short collaboration chains. Combined with an exceptionally high clustering coefficient (0.7400), far exceeding that of a comparable random graph, these findings mathematically validate that the anime industry operates as a highly clustered yet globally efficient Small World network.

These results suggest that the seiyuu ecosystem is sustained by a dense core of influential actors who bridge otherwise distinct production communities, enabling rapid professional interconnectedness across decades of anime history. Beyond identifying central figures, this research provides a quantitative framework for understanding how collaboration structures shape cultural industries.

Future work should expand upon this foundation by incorporating weighted edges based on repeated collaborations, exploring genre- or studio-specific subgraphs, and testing whether the network follows broader scale-free or power-law structural properties. Such analyses would provide even deeper insight into the hierarchy, resilience, and evolution of Japan’s voice-acting industry.

CODE AVAILABILITY

Code is available at <https://github.com/christinec04/Anime-Kevin-Bacon>.

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GENAI DISCLOSURE

Generative AI models such as ChatGPT was used to critique the draft of the paper and copyedit the final version. All code, ideas and analysis are my own.

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